

Github and Environment Setup

Github and Environment Setup

- <https://github.com/csdl-projects/MLCASim.git>
- python 3.11.9 cuda 환경
- mkdir run_scripts (스크립트 넣을 폴더)
- mkdir run (프로그램 돌릴 환경)
- mkdir configs (실험 스크립트 작성할 폴더, yaml 포맷)
- cd run

Data Generation

- HSPICE 환경 셋업
- vi ../run_scripts/data_gen_240801.tcl
 - Parameters
 - -s(d) scanline(dataline) 전체 크기
 - -x(y) scan(data) 방향 sampling rate
 - -v VDH
 - -r resistance
 - -c cap
 - -lr load ratio

≡ data_gen_240801.tcl ×

run_scripts > ≡ data_gen_240801.tcl

```
1  module load license/license
2  module load hspice/2021
3  module load python/3.9.5
4  module list
5
6  #TRAIN
7  python3 ../src/genCircuitData.py -s 64 -d 36 -x 8 -y 6 -v 2.7 -lr 10 &
8  python3 ../src/genCircuitData.py -s 64 -d 36 -x 8 -y 6 -v 2.9 -lr 10 &
9  python3 ../src/genCircuitData.py -s 64 -d 36 -x 8 -y 6 -v 3.1 -lr 10 &
10 python3 ../src/genCircuitData.py -s 64 -d 2 -x 8 -y 6 -v 2.7 -lr 10 &
11 python3 ../src/genCircuitData.py -s 64 -d 2 -x 8 -y 6 -v 2.9 -lr 10 &
12 python3 ../src/genCircuitData.py -s 64 -d 2 -x 8 -y 6 -v 3.1 -lr 10 &
13 python3 ../src/genCircuitData.py -s 2 -d 36 -x 8 -y 6 -v 2.7 -lr 10 &
14 python3 ../src/genCircuitData.py -s 2 -d 36 -x 8 -y 6 -v 2.9 -lr 10 &
15 python3 ../src/genCircuitData.py -s 2 -d 36 -x 8 -y 6 -v 3.1 -lr 10 &
16
17 #TEST
18 python3 ../src/genCircuitData.py -s 64 -d 36 -x 8 -y 6 -v 3.0 -lr 10 &
19 python3 ../src/genCircuitData.py -s 64 -d 36 -x 8 -y 6 -v 4.0 -lr 10 &
```

모델 학습

- vi ../configs/basic.yaml
 - 모델 학습용 yaml 파일 작성

```
! 240805_2_mixer.yaml M ×
configs > ! 240805_2_mixer.yaml
1 project: MLCASim
2 method: grid
3 metric:
4   goal: minimize
5   name: MSE
6 parameters:
7   name:
8     value: '240802_CircuitMixer_exp2'
9   device:
10    value: '3'
11 epoch:
12   value: 5000
13 resume:
14   value: 0
15 verbose:
16   value: 0
17 plot:
18   value: 1
19 test_plot:
20   value: 1
21 model_option:
22   value: 3
23 normalization:
24   value: 'linear'
25 cases:
26   value : [
27     [64, 8, 36, 6, 8.0, 0.008, 2, 3.1, 2], [64, 8, 36, 6, 8.0, 0.008, 2, 2.1],
28     [64, 8, 36, 6, 8.0, 0.008, 2, 3.1, 10], [64, 8, 36, 6, 8.0, 0.008, 2, 2.1],
29     [64, 8, 36, 6, 16.0, 0.008, 2, 3.1, 2], [64, 8, 36, 6, 16.0, 0.008, 2, 2.1],
30     [64, 8, 36, 6, 8.0, 0.016, 2, 3.1, 2], [64, 8, 36, 6, 8.0, 0.016, 2, 2.1],
31   ]
```

```
! 240805_2_mixer.yaml M ×
configs > ! 240805_2_mixer.yaml
6 parameters:
25 cases:
32 test_cases:
33   value : [
34     [64, 8, 36, 6, 8.0, 0.008, 2, 3.0, 2], [64, 8, 36, 6, 16.0, 0.008, 2,
35   ]
36 learning_rate:
37   value: 0.0005
38 type_index:
39   value : 2
40 window_size:
41   value : 10
42 num_layers:
43   value : 4
44 hidden_size:
45   value : 256
46 hidden_channel:
47   value : 64
48 start:
49   value : 0
50 end:
51   value : 200000
52 resolution:
53   value : 0.001
54 batch_size:
55   value : 30
```

모델 train 및 inference

- vi ../configs/basic.yaml

- 모델 학습용 yaml 파일 작성

- Parameters

- name: 모델 base 이름
 - device: gpu 번호
 - epoch: 학습 epoch
 - resume: 학습 재개 (로드)
 - verbose: 로깅
 - plot: 학습 결과 plot (../plot)
 - test_plot: inference 파일
 - model_option: 사용할 모델 번호 (1: simple, 2: IHE-CLSTM, 3: CircuitMixer)
 - cases: 학습에 사용할 spice 정보

```
! 240805_2_mixer.yaml M X
configs > ! 240805_2_mixer.yaml
1 project: MLCASim
2 method: grid
3 metric:
4   goal: minimize
5   name: MSE
6 parameters:
7   name:
8     value: '240802_CircuitMixer_exp2'
9   device:
10    value: '3'
11 epoch:
12   value: 5000
13 resume:
14   value: 0
15 verbose:
16   value: 0
17 plot:
18   value: 1
19 test_plot:
20   value: 1
21 model_option:
22   value: 3
23 normalization:
24   value: 'linear'
25 cases:
26   value: [
27     [64, 8, 36, 6, 8.0, 0.008, 2, 3.1, 2], [64, 8, 36, 6, 8.0, 0.008, 2, 2.
28     [64, 8, 36, 6, 8.0, 0.008, 2, 3.1, 10], [64, 8, 36, 6, 8.0, 0.008, 2, 2
29     [64, 8, 36, 6, 16.0, 0.008, 2, 3.1, 2], [64, 8, 36, 6, 16.0, 0.008, 2,
30     [64, 8, 36, 6, 8.0, 0.016, 2, 3.1, 2], [64, 8, 36, 6, 8.0, 0.016, 2, 2.
31   ]
```

```
! 240805_2_mixer.yaml M X
configs > ! 240805_2_mixer.yaml
6 parameters:
25 cases:
32 test_cases:
33   value: [
34     [64, 8, 36, 6, 8.0, 0.008, 2, 3.0, 2], [64, 8, 36, 6, 16.0, 0.008, 2,
35   ]
36 learning_rate:
37   value: 0.0005
38 type_index:
39   value: 2
40 window_size:
41   value: 10
42 num_layers:
43   value: 4
44 hidden_size:
45   value: 256
46 hidden_channel:
47   value: 64
48 start:
49   value: 0
50 end:
51   value: 200000
52 resolution:
53   value: 0.001
54 batch_size:
55   value: 30
```

모델 train 및 inference

- vi ../configs/basic.yaml
 - 모델 학습용 yaml 파일 작성
 - Model Parameters
 - test cases: inference 에 사용할 spice 정보
 - type_index: 학습 대상 (0: DRG, 1: DRS, 2: Diode)
 - window_size: SPICE 초기 voltage
 - 나머지 ml paramters

```
! 240805_2_mixer.yaml M ×
configs > ! 240805_2_mixer.yaml
1 project: MLCASim
2 method: grid
3 metric:
4   goal: minimize
5   name: MSE
6 parameters:
7   name:
8     value: '240802_CircuitMixer_exp2'
9   device:
10    value: '3'
11 epoch:
12   value: 5000
13 resume:
14   value: 0
15 verbose:
16   value: 0
17 plot:
18   value: 1
19 test_plot:
20   value: 1
21 model_option:
22   value: 3
23 normalization:
24   value: 'linear'
25 cases:
26   value: [
27     [64, 8, 36, 6, 8.0, 0.008, 2, 3.1, 2], [64, 8, 36, 6, 8.0, 0.008, 2, 2.
28     [64, 8, 36, 6, 8.0, 0.008, 2, 3.1, 10], [64, 8, 36, 6, 8.0, 0.008, 2, 2
29     [64, 8, 36, 6, 16.0, 0.008, 2, 3.1, 2], [64, 8, 36, 6, 16.0, 0.008, 2,
30     [64, 8, 36, 6, 8.0, 0.016, 2, 3.1, 2], [64, 8, 36, 6, 8.0, 0.016, 2, 2.
31   ]
```

```
! 240805_2_mixer.yaml M ×
configs > ! 240805_2_mixer.yaml
6 parameters:
25 cases:
32 test_cases:
33   value: [
34     [64, 8, 36, 6, 8.0, 0.008, 2, 3.0, 2], [64, 8, 36, 6, 16.0, 0.008, 2,
35   ]
36 learning_rate:
37   value: 0.0005
38 type_index:
39   value: 2
40 window_size:
41   value: 10
42 num_layers:
43   value: 4
44 hidden_size:
45   value: 256
46 hidden_channel:
47   value: 64
48 start:
49   value: 0
50 end:
51   value: 200000
52 resolution:
53   value: 0.001
54 batch_size:
55   value: 30
```

모델 train 및 inference

- ../run_scripts/
 - 모델 학습용 스크립트 파일 작성
 - Parameters
 - -c: config 파일 이름 (.yaml 빼고)
 - --config_dir: config 파일 위치 (기본 ../configs)

≡ process_240807_2_mixer.tcl U ✕

run_scripts > ≡ process_240807_2_mixer.tcl

1 module load python/3.9.5

2 python3 ../src/main.py -c 240807_2_mixer --config_dir ../configs

모델 inference

- ../run_scripts/
 - 모델 학습용 스크립트 파일 작성
 - Parameters
 - -c: config 파일 이름 (.yaml 빼고)
 - --config_dir: config 파일 위치 (기본 ../configs)

≡ *inference_240807_2_mixer.tcl* U X

run_scripts > ≡ inference_240807_2_mixer.tcl

1 module load python/3.9.5

2 python3 ../src/inference.py -c 240807_2_mixer